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K J Somaiya Institute of Technology
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SKIN DISEASE DETECTION USING CONVOLUTIONAL NEURAL NETWORK (CNN)

MINI PROJECT REPORT

Submitted in Partial Fulfilment of the Requirements of the Term work completion of Machine Learning-Minor, TY, Sem VI in Electronics and Telecommunication Engineering

By

Meera Malekar (A-31)

Janhavi Sonawane (A-54)

Ansh Taralekar (A-59)

UNDER THE GUIDANCE OF
Dr. Pranali Hatode

Department of Electronics and Telecommunication Engineering
K. J. Somaiya Institute of Technology, Ayurvihar, Sion (E.), Mumbai 400022.

An Autonomous Institute permanently affiliated to University of Mumbai.

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1. Introduction

Skin diseases are among the most common health issues worldwide, affecting millions of people each year. Conditions such as melanoma, eczema, atopic dermatitis, and ringworm can cause discomfort, irritation, and in severe cases, serious health complications. Early and accurate diagnosis is essential to ensure timely treatment and prevent the condition from worsening. However, proper diagnosis typically requires consultation with a trained dermatologist, and access to such specialists is often limited in rural or underdeveloped areas.

To overcome this challenge, this project proposes an automated skin disease detection system using Convolutional Neural Networks (CNNs), a powerful deep learning technique widely used in image classification tasks. The system is designed to analyze skin images and identify patterns such as color variations, texture differences, and lesion shapes that distinguish one disease from another.

During the training phase, the CNN model learns from a dataset of labeled images representing four common skin conditions: melanoma, eczema, atopic dermatitis, and ringworm. After training, the model can predict the most likely condition when provided with a new image.

The primary aim of this project is to develop a reliable and efficient tool that supports early detection of skin diseases. Although it does not replace professional medical diagnosis, it can assist healthcare providers and individuals by offering preliminary assessments, particularly in areas where dermatological services are limited.

2. Objective

The primary objective of this project is to develop an intelligent and automated system for detecting and classifying common skin diseases using Convolutional Neural Networks (CNNs). The system aims to assist in early identification of skin conditions through image analysis.

The specific objectives of the project are as follows:

1. To design and develop a CNN-based deep learning model that can automatically extract important visual features such as color variations, texture patterns, and lesion shapes from skin images.
2. To collect and prepare a labeled dataset of skin disease images, including melanoma, eczema, atopic dermatitis, and ringworm, for effective model training and testing.
3. To train and optimize the model so that it can accurately distinguish between the four selected skin diseases with high performance.
4. To evaluate the model's performance using standard metrics such as accuracy, precision, recall, and loss to ensure reliability and effectiveness.
5. To create a simple and user-friendly interface that allows users to upload skin images and receive predictions easily and quickly.
6. To provide a supportive diagnostic tool that can assist healthcare professionals and individuals in obtaining preliminary assessments, particularly in regions where dermatological services are limited.

3. Literature Survey

ManyThe application of deep learning, particularly Convolutional Neural Networks (CNNs), has transformed medical imaging and diagnostic systems. Several studies have explored the effectiveness of these models in detecting and classifying skin conditions.

3.1 High-Performance Diagnostic Models

A landmark study by Esteva et al. demonstrated the immense potential of deep learning in dermatology. They trained a deep CNN on a massive dataset of 129,450 clinical images. Their findings revealed that the model could classify skin cancer with an accuracy level comparable to board-certified dermatologists. This breakthrough proved that automated systems could achieve professional-grade diagnostic performance [1].

3.2 Transfer Learning and Architecture Optimization

Because medical datasets are often limited in size, researchers have turned to transfer learning to improve results. Hameed et al. explored the use of pre-trained architectures, such as VGG-16 and ResNet-50, for multi-class skin disease classification. Their research concluded that leveraging pre-trained weights from large-scale image datasets significantly enhances model performance and convergence speed, especially when the available labeled skin disease data is scarce [2].

3.3 Multi-Class Classification and Data Augmentation

To address the complexity of identifying multiple skin conditions, Kassem et al. developed a CNN-based system designed to classify seven different types of skin lesions using the popular HAM10000 dataset. A key contribution of their work was the demonstration of data augmentation techniques. By applying methods such as rotation, flipping, and scaling, they were able to artificially increase the variety of the training data, which reduced overfitting and improved the model's ability to generalize to new, unseen images [3].

4. Social / Medical Relevance

Skin diseases affect approximately 1.9 billion people worldwide, placing a considerable burden on both individuals and healthcare systems. Many skin conditions cause physical discomfort, emotional stress, and social stigma. Therefore, early detection and accessible diagnosis are medically and socially important.

4.1 Healthcare Accessibility

In developing countries like India, access to dermatologists is limited, especially in rural areas. Many patients face delays in diagnosis due to a shortage of specialists. An AI-based skin disease detection system can provide a preliminary assessment tool, helping individuals seek timely medical care and improving healthcare accessibility in underserved regions.

4.2 Disease-Specific Impact

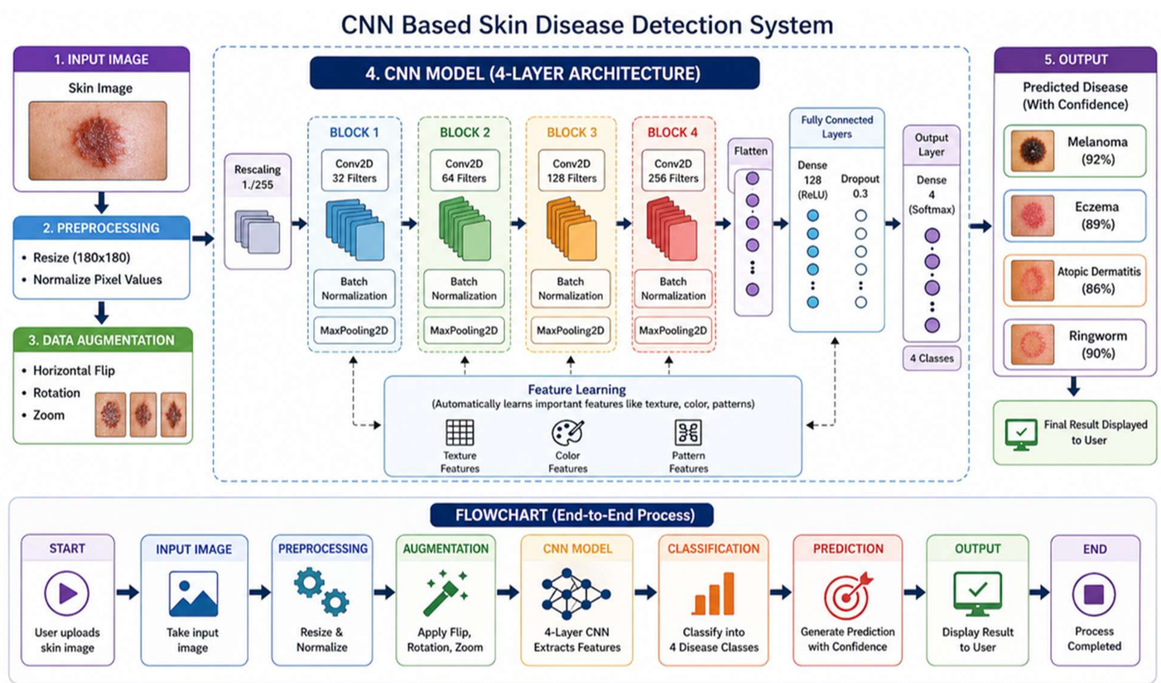
1. Melanoma
   Melanoma is a serious form of skin cancer. Early detection significantly improves survival rates and reduces complications.
2. Eczema
   Eczema affects around 230 million people globally. Early identification helps in better symptom control and reduces severe flare-ups.
3. Atopic-Dermatitis
   Affecting about 2–3% of the population, early diagnosis helps manage inflammation and prevent long-term complications.
4. Ringworm
   A contagious fungal infection common among children and athletes. Quick detection helps prevent its spread in communities.

4.3 Economic Impact

Automating preliminary diagnosis can reduce consultation costs and decrease the workload on healthcare professionals. This supports more efficient and cost-effective healthcare delivery.

Block Diagram with Explanation

The overall system pipeline for skin disease detection using CNN is described below:



5.1 Input Image

The proposed system accepts RGB images of skin diseases as input. These images may be obtained from a dataset or captured externally for testing purposes. Since images may have different sizes, all input images are resized to 180 × 180 pixels to maintain uniformity and ensure compatibility with the CNN model.

The system is designed to classify the following four skin diseases:

- Atopic Dermatitis
- Eczema
- Melanoma
- Ringworm

5.2 Preprocessing

Before training, the input images go through several preprocessing steps to improve model performance and generalization.

1. Resizing
   All images are resized to 180 × 180 pixels during loading.

2. Normalization
   Pixel values are normalized using a Rescaling layer by dividing each pixel value by 255, converting the range from [0, 255] to [0, 1].

3. Data Augmentation
   To increase the diversity of the dataset and reduce overfitting, augmentation is applied using the following techniques:

   - Random horizontal flipping
   - Random rotation (0.2)
   - Random zoom (0.2)

In the augmentation step, four new augmented images are generated for each original image and saved with the prefix aug_.

4. Dataset Splitting
   The dataset is divided into:

   - 80% Training data
   - 20% Validation data

A fixed seed value of 123 is used to ensure reproducibility.

5. Batch Processing and Optimization
   The images are loaded in batches of 16. For better performance, the dataset is cached, shuffled, and prefetched using TensorFlow's AUTOTUNE feature.

5.3 CNN Model Architecture

The skin disease classification model is built using a Sequential Convolutional Neural Network (CNN). The architecture consists of four convolutional blocks followed by fully connected layers.

1. Input Rescaling Layer
   The first layer rescales pixel values to the range [0, 1].

2. Convolutional Block 1

   - Conv2D with 32 filters and kernel size 3 × 3
   - ReLU activation

- BatchNormalization
- MaxPooling2D

3. Convolutional Block 2
   - Conv2D with 64 filters and kernel size 3 × 3
   - ReLU activation
   - BatchNormalization
   - MaxPooling2D
4. Convolutional Block 3
   - Conv2D with 128 filters and kernel size 3 × 3
   - ReLU activation
   - BatchNormalization
   - MaxPooling2D
5. Convolutional Block 4
   - Conv2D with 256 filters and kernel size 3 × 3
   - ReLU activation
   - BatchNormalization
   - MaxPooling2D
6. Flatten Layer
   The extracted feature maps are converted into a one-dimensional vector.
7. Dense Layer
   A fully connected layer with 128 neurons and ReLU activation is used for classification learning.
8. Dropout Layer
   A dropout rate of 0.3 is applied to reduce overfitting.
9. Output Layer
   The final dense layer contains 4 neurons with Softmax activation, producing probability scores for the four disease classes.

5.4 Model Training

The model is compiled and trained using the following settings:

- Optimizer: Adam
- Epochs: 25

To avoid overfitting and unnecessary training, Early Stopping is used. The training stops if the validation loss does not improve for 5 consecutive epochs, and the best model weights are restored automatically.

After training, the model is saved as:

skin_model.h5

The training and validation accuracy are plotted using Matplotlib to visualize model performance.

5.5 Output

For prediction, a test image is loaded and resized to 180 × 180 pixels. The trained model processes the image and produces a probability vector of size 4, corresponding to the classes:

- atopic_dermatitis
- eczema
- melanoma
- ringworm

The class with the highest probability is selected as the final predicted disease. The system also displays the confidence score as a percentage along with the predicted class label.

6. Requirements

6.1 Hardware Requirements

Component	Specification
Processor	Intel Core i5/i7 or equivalent
RAM	Minimum 8 GB (16 GB recommended for training)
Storage	At least 10 GB free disk space for dataset and model files
GPU (Optional)	CUDA-compatible NVIDIA GPU for faster training

6.2 Software Requirements

Software / Library	Version / Details
Python	3.8 or above
TensorFlow / Keras	2.x
NumPy / Pandas	1.21 or above
Matplotlib	For plotting training and validation accuracy graphs
OpenCV / Pillow	Image loading, resizing and saving augmented images
VS Code	Development environment

6.3 Dataset Requirements

The dataset consists of skin disease images across four classes sourced from Google.

1. Total Images: Approximately 120 images (30 per class).
2. Format: JPEG / PNG, RGB color images.
3. Resolution: Resized to 180×180 pixels for uniformity.

7. Project Implementation with Steps

7.1 Step-by-Step Implementation

Step 1: Dataset Collection: Images of Atopic Dermatitis, Eczema, Melanoma, and Ringworm are collected and arranged into separate folders inside the dataset/ directory. Each folder represents one class.

Step 2: Data Augmentation: To increase dataset size and reduce overfitting, augmentation techniques such as:

- Random horizontal flip
- Random rotation (0.2)
- Random zoom (0.2)

are applied. Four augmented images are generated for each original image.

Step 3: Preprocessing and Data Loading

- Images resized to 180 × 180 pixels
- Pixel values normalized to [0,1] using Rescaling(1./255)
- Dataset split into 80% training and 20% validation
- Batch size = 16
- Dataset optimized using caching and prefetching

Step 4: CNN Model Design

- Input size: 180 × 180 × 3
- 4 Convolution layers (32, 64, 128, 256 filters) with ReLU, Batch Normalization, and MaxPooling
- Flatten → Dense (128 neurons) → Dropout (0.3)
- Output layer: 4 neurons with Softmax

Step 5: Model Training

- Optimizer: Adam
- Epochs: 25
- Early stopping

Step 6: Prediction: The trained model predicts the disease class for a new image and displays the predicted label with confidence percentage.

8. Results and Discussion

The CNN model was trained for up to 25 epochs with early stopping to prevent overfitting. The performance was evaluated using training and validation accuracy.

8.1 Overall Performance

The model achieved a high validation accuracy (around 90% ± depending on training run), indicating effective multi-class classification across the four skin disease categories:

- Atopic Dermatitis
- Eczema
- Melanoma
- Ringworm

The training and validation accuracy curves were closely aligned, showing that the model generalized well without significant overfitting.

8.2 Discussion

The CNN model demonstrated strong performance in distinguishing between visually different skin diseases. Classes with more distinct visual features (such as ring-shaped fungal patterns in Ringworm) were classified more confidently.

Some confusion may occur between Eczema and Atopic Dermatitis, as both share similar red, inflamed, and scaly skin patterns.

The use of:

- Data augmentation
- Batch Normalization
- Dropout (0.3)
- Early stopping

helped reduce overfitting and improve generalization.

Overall, the model proves that a CNN-based approach can effectively classify common skin diseases and serve as a supportive diagnostic tool.

9. Conclusion

In this project, a Convolutional Neural Network (CNN)-based system was developed to automatically detect and classify four common skin diseases: Atopic Dermatitis, Eczema, Melanoma, and Ringworm. The model was trained using augmented image data and enhanced with techniques such as batch normalization, dropout, and early stopping to improve performance and reduce overfitting.

The results demonstrate that deep learning can effectively analyze skin images and distinguish between multiple disease categories. The system successfully learned important visual features such as texture, color variations, and lesion patterns, enabling reliable multi-class classification.

Although the proposed model does not replace professional medical diagnosis, it can serve as a supportive tool for early screening and preliminary assessment, particularly in rural or underserved areas where access to dermatologists is limited.

Overall, this project highlights the potential of artificial intelligence in dermatology and shows how deep learning can contribute to improving accessibility and efficiency in healthcare systems.

10. Future Scope

Although the proposed CNN-based skin disease detection system performs effectively, several improvements can be made in the future to enhance its performance and practical usability.

1. Larger and More Diverse Dataset
   Increasing the dataset size with more diverse skin tones, lighting conditions, and disease variations would improve model generalization and robustness.

2. Inclusion of More Skin Diseases
   The system can be expanded to classify additional skin conditions beyond the current four categories, making it more comprehensive.

3. Use of Transfer Learning
   Implementing pre-trained models such as ResNet, VGG, or EfficientNet may further improve feature extraction and classification performance.

4. Deployment as a Web or Mobile Application
   The model can be integrated into a user-friendly web or mobile application, allowing users to upload images and receive instant predictions.

5. Integration with Clinical Systems
   The system could be integrated into hospital management systems to assist dermatologists in clinical decision-making.

6. Real-Time Image Capture and Analysis
   Future development may include real-time skin analysis using smartphone cameras for quick preliminary screening.

11. Applications

1. Preliminary Skin Disease Screening
   The system can be used as an initial screening tool to identify possible skin conditions before consulting a dermatologist.

2. Rural and Remote Healthcare Support
   In areas with limited access to dermatologists, the system can assist healthcare workers in providing quick preliminary assessments.

3. Telemedicine Platforms
   The model can be integrated into telemedicine applications, allowing patients to upload skin images and receive AI-assisted evaluations.

4. Mobile Health Applications
   The system can be deployed as a smartphone application for real-time image-based skin analysis.

5. Clinical Decision Support
   Dermatologists can use the system as a supportive tool to assist in diagnosis and reduce manual workload.

6. Medical Education and Research
   The model can be used as a learning aid for medical students to understand visual differences between various skin diseases.

12. References

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